

# Translatability FFGFT $\leftrightarrow$

## Hilbert-space QM

Bijjective bridge between mode formalism and  
standard quantum mechanics

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May 2026

### Preamble

This document formulates the **complete translatability** between the FFGFT mode formalism and standard Hilbert-space quantum mechanics. The translation is not an approximation or specialisation, but a bijective structural statement: every QM calculation can be mapped into FFGFT language, and conversely every FFGFT statement about quantum systems is reproducible in Hilbert-space language.

**This is methodologically important:** it means that FFGFT does not *replace* quantum mechanics, but *extends* it. QM is a subset of FFGFT (see Doc. 202 §22). Practitioners familiar with Hilbert-space language can use FFGFT without changing their tools – the bridge is the translation table below.

### Where we are, geometrically

Before the formal translation tables, a brief narrative picture of the underlying geometry. This matters because we are about to work with *cylindrical coordinates* – and

the cylinder is not the full FFGFT geometry, but a convenient reduction.

**The full FFGFT geometry is four-dimensional.** The fundamental space on which all FFGFT statements live is the 4-torus  $T^4 = T^3 \times S^1$  – three spatial winding directions and one temporal winding (Doc. 202 §13, Doc. 210). The wave function of a mode  $\varphi_{n,l,j}$  is a function on this 4D torus. From this geometry follow all masses, the fine-structure constant, cosmology – everything that distinguishes FFGFT from pure quantum mechanics.

**For a single qubit, less suffices.** When we want to describe just a single qubit – a two-level system at fixed time, without mass dynamics – most of these winding directions are not active. Two rotational degrees of freedom remain: one encoding “where between  $|0\rangle$  and  $|1\rangle$  we are” (this becomes our  $z$ ), and one for the phase (this becomes our  $\theta$ ). The natural space for these two rotational degrees of freedom is a 2-torus  $T^2 = S^1 \times S^1$ .

**From 2-torus to cylinder.** If we now assume that one major radius  $R$  of the 2-torus is very large compared to the tube radius  $r$ , then the local geometry is no longer curved – it looks like a cylinder. The cylinder  $\mathbb{R} \times S^1$  is thus the limit  $R \rightarrow \infty$  of the 2-torus. Topologically one loop is “cut open”: from two circles we get a line plus a circle. In practice this means: in cylindrical coordinates  $(z, r, \theta)$ ,  $z$  is a *linear* axis (from  $-1$  to  $+1$ ), no longer cyclic.

**From cylinder to Bloch sphere.** If we contract the cylinder top and bottom to a point – the two “poles” – we obtain a sphere  $S^2$ . This is the famous **Bloch sphere**

of standard quantum mechanics. It is the second, further reduction: cylinder with merged ends.

**Hierarchy of reductions.**

| Geometry                         | Dim. | Use                                                |
|----------------------------------|------|----------------------------------------------------|
| 4-torus $T^4$                    | 4    | full FFGFT (masses, $\alpha$ , cosmology)          |
| 3-torus $T^3$                    | 3    | mode space at fixed time                           |
| 2-torus $T^2$                    | 2    | single qubit, without reduction                    |
| Cylinder $\mathbb{R} \times S^1$ | 2    | limit $R \rightarrow \infty$ , local approximation |
| Bloch sphere $S^2$               | 2    | poles contracted, standard QM                      |

**What the reductions cost.** Each step of reduction loses something. From the 4-torus to the 3-torus one loses the time winding – and with it all mass statements (Doc. 210). From the 3-torus to the 2-torus one loses a spatial winding – and with it the connection to the third lepton generation and many spin structures. From the 2-torus to the cylinder one loses a *compact loop* – this is the weakest reduction, and exactly here sits the **fractal correction factor**  $K_{\text{frak}} \approx 0.9867$  that appears in the bit-flip rotations of Doc. 175.  $K_{\text{frak}}$  is the “memory” of the cylinder of its toroidal origin.

**What we translate here.** The translation tables below apply to the *qubit sector*, i.e. the lower three rows of the hierarchy – 2-torus, cylinder, Bloch sphere. Standard

quantum mechanics works on the Bloch sphere; FFGFT works on the cylinder (or the 2-torus, if one takes  $K_{\text{frak}}$  seriously). The translation between these levels is lossless, provided one carries the small geometric corrections ( $K_{\text{frak}}$ ,  $\Delta\text{CHSH}$ ) along.

The full FFGFT geometry – the 4-torus – lies one step higher and provides what Hilbert-space QM cannot provide: the values of the input parameters (masses,  $\alpha$ ,  $\xi$ ). This is discussed in §10 below.

## 1 States: $(z, r, \theta)$ vs. $(\alpha, \beta)$

### 1.1 Hilbert-space representation

In standard QM a qubit is a vector in  $\mathbb{C}^2$ :

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle, \quad |\alpha|^2 + |\beta|^2 = 1, \quad \alpha, \beta \in \mathbb{C}. \quad (1)$$

With global phase freedom this is a point on the Bloch sphere (Hopf fibration  $S^3 \rightarrow S^2$ ).

### 1.2 FFGFT representation

In FFGFT the same qubit is a point  $(z, r, \theta)$  on the cylinder/torus:

$$z \in [-1, 1], \quad r \in [0, 1], \quad \theta \in [0, 2\pi), \quad z^2 + r^2 = 1. \quad (2)$$

$z$  is the projection onto the computational basis axis,  $r$  the radial superposition amplitude,  $\theta$  the phase (Doc. 147 §1.2).

### 1.3 Bijective translation

The two representations are linked by:

$$\boxed{\alpha = \sqrt{\frac{1+z}{2}} \cdot e^{i\theta/2}, \quad \beta = \sqrt{\frac{1-z}{2}} \cdot e^{-i\theta/2}.} \quad (3)$$

Conversely:

$$z = |\alpha|^2 - |\beta|^2, \quad r = 2|\alpha\beta|, \quad \theta = \arg(\alpha) - \arg(\beta). \quad (4)$$

### Consistency checks:

- $|\alpha|^2 + |\beta|^2 = \frac{1+z}{2} + \frac{1-z}{2} = 1. \quad \checkmark$
- $z^2 + r^2 = (|\alpha|^2 - |\beta|^2)^2 + 4|\alpha|^2|\beta|^2 = 1. \quad \checkmark$
- $|0\rangle$  corresponds to  $(z, r, \theta) = (1, 0, *)$ .  $\checkmark$
- $|1\rangle$  corresponds to  $(z, r, \theta) = (-1, 0, *)$ .  $\checkmark$

**Conceptual difference:** in QM,  $|\alpha|^2$  is a *probability* (Born rule). In FFGFT,  $r^2 = 1 - z^2$  is an *energy density* of the radial configuration. The two quantities are *numerically equal* at large statistics (Doc. 147), but conceptually different: FFGFT is a deterministic geometric model whose statistical predictions coincide with the Born rule.

## 2 Operators: Pauli matrices vs. rotations

### 2.1 Hilbert-space operators

The three Pauli matrices generate all single-qubit operations:

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \quad (5)$$

Algebra:  $\sigma_i \sigma_j = \delta_{ij} \mathbb{1} + i\epsilon_{ijk} \sigma_k$ .

### 2.2 FFGFT rotations

On the Bloch cylinder, the three Pauli matrices correspond to three geometric rotations (Doc. 175 §3):

- $\sigma_z$  – rotation about the cylinder axis (phase rotation  $\theta \rightarrow \theta + \pi$ ).
- $\sigma_x$  – 2D rotation in the  $(z, r)$  plane by angle  $\pi$  (bit flip).
- $\sigma_y$  – orthogonal 2D rotation, combining  $\sigma_x$  and phase rotation.

## 2.3 Translation table

|            | Operator Hilbert-space matrix                                      | FFGFT geometry                                                      |
|------------|--------------------------------------------------------------------|---------------------------------------------------------------------|
| $\sigma_z$ | $\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$                    | Phase rotation<br>$\theta \rightarrow \theta + \pi$                 |
| $\sigma_x$ | $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$                     | $(z, r)$ -rotation by<br>$\pi K_{\text{frak}}$                      |
| $\sigma_y$ | $\begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$                    | Rotation with<br>phase tilt                                         |
| $H$        | $\frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$ | Swap $z \leftrightarrow r$ ,<br>$\theta \rightarrow \theta + \pi/2$ |
| $T$        | $\begin{pmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \end{pmatrix}$            | Phase rotation<br>$\theta \rightarrow \theta + \pi/4$               |

**The only FFGFT-specific deviation** is the factor  $K_{\text{frak}} \approx 0.9867$  in the  $\sigma_x$  rotation (Doc. 175). This deviation follows from the fractal spacetime dimension  $D_f = 2.999867$  and is a *testable prediction*: all  $\pi$ -gates deviate systematically by  $\Delta\alpha \approx 0.013\%$  from the ideal value. In Hilbert-space formalism this would be an ad hoc correction modeled as hardware noise; in FFGFT it follows from the geometry.

## 3 Quantum gates: complete translatability

### 3.1 Single-qubit gates

Every unitary single-qubit gate  $U \in U(2)$  can be represented as a Bloch-sphere rotation:

$$U = e^{i\alpha} R_{\hat{n}}(\phi) = e^{i\alpha} (\cos(\phi/2) \mathbb{1} - i \sin(\phi/2) \hat{n} \cdot \vec{\sigma}). \quad (6)$$

In FFGFT this is a rotation of the  $(z, r, \theta)$  point about axis  $\hat{n}$  by angle  $\phi$ . The translation is exact:  $U|\psi_{\text{QM}}\rangle$  corresponds to the rotation of the FFGFT cylinder point about  $\hat{n}$  by  $\phi$ .

### 3.2 Two-qubit gates

The CNOT gate:

$$\text{CNOT} = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \end{pmatrix} \quad (7)$$

translates in FFGFT as: *if  $z_A = -1$  (control qubit on  $|1\rangle$ ), perform a bit flip on qubit B*. This is geometric conditioning: the rotation of one cylinder point depends on the  $z$  value of the other. Topologically this corresponds to a linkage of the two cylinders (entanglement).

### 3.3 Universality

The set  $\{H, T, \text{CNOT}\}$  is universal for QM (Solovay-Kitaev theorem). Since each of these gates is representable as a geometric operation in FFGFT, FFGFT is *fully universal for quantum computation*. There is no QM statement that cannot be formulated in FFGFT.

## 4 Tensor products and many-qubit spaces

### 4.1 Hilbert-space construction

For  $n$  qubits the Hilbert space is  $\mathcal{H}_n = \bigotimes_{i=1}^n \mathbb{C}^2$  with dimension  $2^n$ . A general state is:

$$|\Psi\rangle = \sum_{x \in \{0,1\}^n} \alpha_x |x\rangle, \quad \sum_x |\alpha_x|^2 = 1. \quad (8)$$

### 4.2 FFGFT product space

In FFGFT  $n$  qubits correspond to  $n$  independent mode configurations  $(z_i, r_i, \theta_i)$  on a shared torus geometry. Entan-

lement is represented as a *topological linkage*: entangled qubits are nodes of the mode configuration that cannot be reduced to product form (Doc. 175 §4).

### 4.3 Bell states

The maximally entangled Bell state

$$|\Phi^+\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle) \quad (9)$$

has in FFGFT a geometric characterisation as a *topologically linked cylinder configuration*, in which the individual  $(z_i, r_i, \theta_i)$  values are not independent.

**CHSH prediction:** standard QM gives  $|\langle \text{CHSH} \rangle| \leq 2\sqrt{2}$  (Tsirelson bound). FFGFT predicts a small, geometrically motivated deviation:

$$\Delta \text{CHSH} \approx 10^{-5} \quad (\text{geometric, from } \xi \text{ and } K_{\text{frak}}). \quad (10)$$

This deviation lies below the current hardware noise floor and is an open experimental task for high-precision Bell tests (Doc. 175 §6, Doc. 023).

### 4.4 Exponential growth

Both formalisms reproduce the exponential growth  $\dim = 2^n$ . In Hilbert space through tensor product; in FFGFT through  $n$  independent mode configurations with topological entanglements. The number of real parameters per state is in both cases  $2 \cdot 2^n - 2$ .

## 5 Measurement

### 5.1 Hilbert-space Born rule

A measurement in the computational basis yields outcome 0 with probability  $|\alpha|^2$  and outcome 1 with probability  $|\beta|^2$ . The state collapses to  $|0\rangle$  or  $|1\rangle$  respectively.



## 5.2 FFGFT measurement as polar transition

In FFGFT, measurement is *not* a probabilistic collapse, but a *geometric interaction* that drives the  $(z, r, \theta)$  point to the nearest stable extremum ( $z = +1$  or  $z = -1$ ). This is a deterministic motion in mode space – but due to the unknown phase  $\theta$  and tiny variations in initial conditions, it is *statistically* indistinguishable from Born-rule behaviour.

## 5.3 Numerical equivalence

At large statistics (many repetitions), the FFGFT measurement distribution converges to the Born rule:

$$P_{\text{FFGFT}}(z = +1) = \frac{1+z}{2} = |\alpha|^2 = P_{\text{QM}}(0). \quad (11)$$

The two formalisms yield *identical* statistical predictions.

**Conceptual difference:** the Bell-test realism debate resolves geometrically in FFGFT, since entanglement is represented as a topological link – not as nonlocal correlation. The experimental predictions are the same.

# 6 Unitary evolution: Schrödinger equation

## 6.1 Hilbert-space Schrödinger

The time evolution in QM is:

$$i\hbar \frac{\partial |\psi\rangle}{\partial t} = H|\psi\rangle. \quad (12)$$

## 6.2 FFGFT mode evolution

In FFGFT the time evolution is the deterministic motion of the mode configuration  $\varphi_{n,l,j}$  on the torus  $T^3 \times \mathbb{R}$  (Doc. 202 §13). The corresponding equation of motion – the FFGFT mode equation – reduces in the low-energy limit ( $E \ll E_\xi$ )

to the Schrödinger equation:

$$i\hbar \frac{\partial \varphi_{n,l,j}}{\partial t} \xrightarrow{E \ll E_\xi} -\frac{\hbar^2}{2m} \nabla^2 \varphi_{n,l,j} + V \varphi_{n,l,j}. \quad (13)$$

At higher energies,  $\xi$  corrections appear, which in the Hilbert-space formalism manifest as Hamiltonian modifications.

## 7 What is fully translatable

### All Hilbert-space statements map into FFGFT:

- States, operators, gates (with or without  $K_{\text{frak}}$  correction),
- Tensor products, entanglement, Bell states, GHZ states,
- Measurement (statistical predictions identical to Born rule),
- Unitary evolution (Schrödinger in low-energy limit),
- Algorithms (Shor, Grover, VQE, etc.).

### All FFGFT QM statements map into Hilbert space:

- The  $(z, r, \theta)$  configuration corresponds uniquely to the  $(\alpha, \beta)$  vector (bridge (3)),
- Geometric rotations correspond to unitary operators,
- Topological entanglements correspond to entangled Hilbert-space vectors.

**Consequence:** every computation that can be performed in one formalism can be performed in the other – with identical end result. Practitioners can choose the formalism familiar to them.

## 8 What is FFGFT-specific and not representable in pure Hilbert space

The translatability holds for *statements about quantum systems*. It does not hold for *statements about the input values* of quantum mechanics:

- The geometric parameter  $\xi_0 = 4/30000$  appears in the Hilbert-space formalism only as an external input. In FFGFT it follows from Higgs matching  $\xi_0 = \lambda_h^2 v^2 / (16\pi^3 m_h^2)$  (Doc. 006).
- The fermion masses  $m_e, m_\mu, m_\tau, m_u, \dots$  appear in Hilbert space as external Yukawa couplings. In FFGFT they all follow parameter-free from  $m_i = r_i \xi_0^{p_i} v$  (Doc. 006).
- The fine-structure constant  $\alpha = \xi_0 \cdot E_0^2$  is derivable in FFGFT, an input in Hilbert space.
- The fractal correction factor  $K_{\text{frak}} = 1 - 100\xi_0 \approx 0.9867$  appears in FFGFT gates as a geometric deviation. In Hilbert space it would have to be modeled as hardware noise.

### This is the structural asymmetry:

- Hilbert space: has all tools for QM calculations, but no explanation for the input values (masses, couplings,  $\alpha$ ).
- FFGFT: provides both the tools for QM calculations *and* the input values from a single geometry.

In this sense QM is a subset of FFGFT (see Doc. 202 §22): QM is FFGFT, restricted to the question *how systems behave*, without the question *why the constants take these values*.

## 9 Consequence: extension, not replacement

FFGFT does not replace quantum mechanics. It extends it to a level where the input values themselves follow from geometry.

### For practitioners this means:

- Those who calculate with Hilbert-space tools can continue to do so. The translation (3) and the operator table allow switching to the FFGFT formalism at any time, without loss.
- Those who calculate with the FFGFT mode formulation additionally have access to masses, couplings, and  $\alpha$  from the same geometry.
- Both languages are *interoperable*. This is precisely the “translatability without contradiction” from Doc. 202 §22: other frameworks can map FFGFT statements into their language, provided they use the translation table.

**Methodological clarification:** FFGFT does not claim to be the only correct representation of quantum mechanics. It is *one* representation that, in addition to the standard QM statements, provides the values of the fundamental constants. Those who do not need this added value can ignore FFGFT – the standard QM statements remain valid.

## 10 Summary

The translation table FFGFT  $\leftrightarrow$  Hilbert-space QM is complete:

1. **States:** bijective bridge  $\alpha = \sqrt{(1+z)/2} e^{i\theta/2}$ ,  $\beta = \sqrt{(1-z)/2} e^{-i\theta/2}$ .
2. **Operators:** Pauli matrices  $\leftrightarrow$  Bloch rotations, with FFGFT correction  $K_{\text{frak}}$  in  $\sigma_x$ .

3. **Gates:** all standard gates ( $H$ ,  $T$ , CNOT, ...) in FFGFT as geometric transformations.
4. **Tensor products:**  $2^n$  growth reproduced, entanglement as topological linkage.
5. **Measurement:** statistically identical to Born rule, conceptually deterministic geometry.
6. **Evolution:** Schrödinger equation as low-energy limit of FFGFT mode evolution.

### **FFGFT added value beyond Hilbert space:**

- parameter-free derivation of  $\alpha$ ,  $m_i$ ,  $\xi$  from a single geometry,
- geometric explanation of the fractal correction factor  $K_{\text{frak}}$  in  $\pi$ -gates (testable in high-precision quantum gate experiments),
- geometric resolution of the Bell-test realism debate (entanglement as topology, not as non-locality),
- small, geometrically motivated CHSH deviation  $\sim 10^{-5}$  as a testable prediction.

**Methodological position:** FFGFT confirms quantum mechanics and extends it. It refutes nothing, but adds structural explanations for the input values. The translatability  $\leftrightarrow$  Hilbert space is a *consistency proof*: any internally consistent theory should be expressible in multiple mathematical languages, and the existence of such translations is evidence of structural soundness.